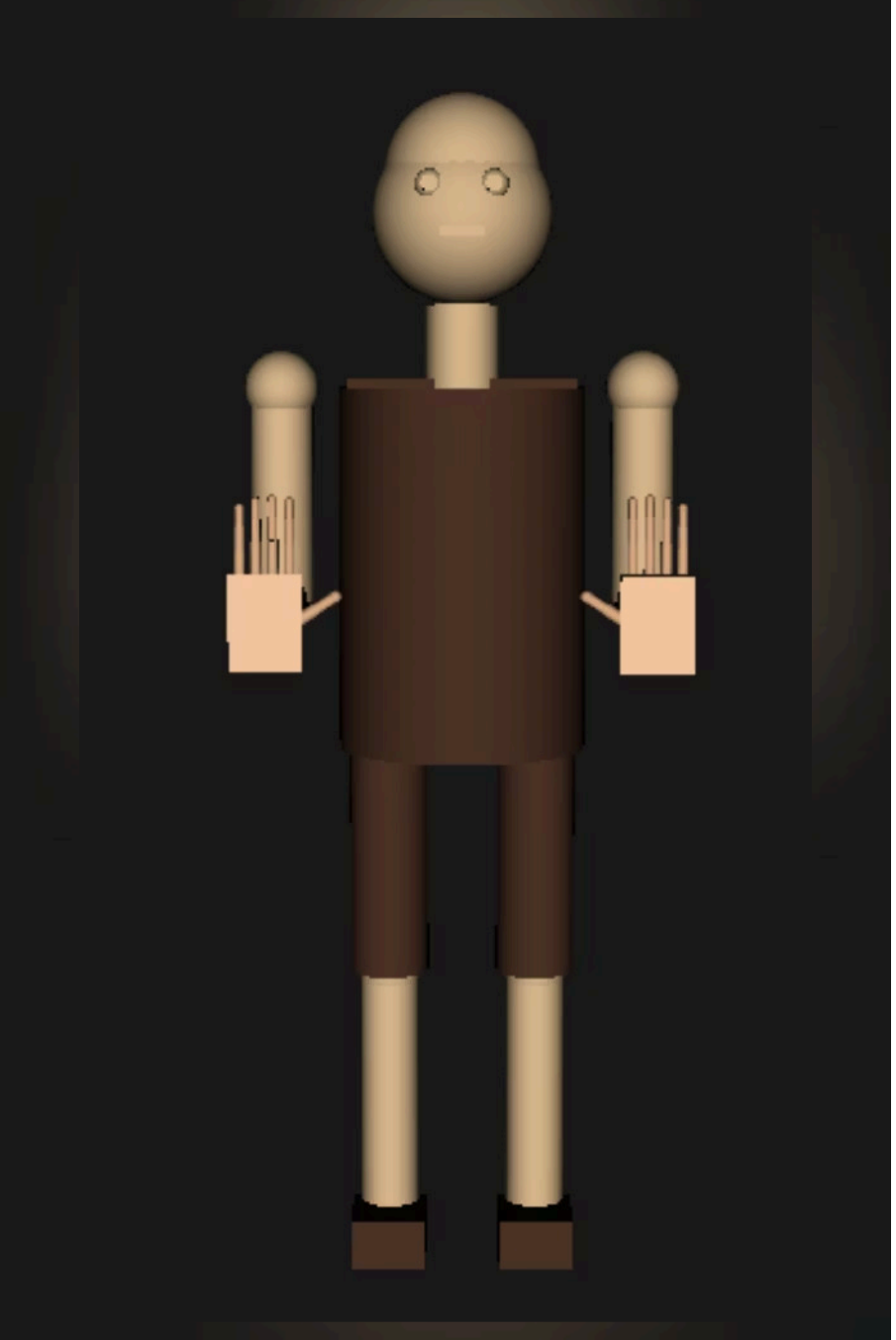

ARM Simulation: ROS 2 Based Robotic Hands and Humanoid System for ASL

A complete robotic simulation system built using ROS 2 that performs American Sign Language using dual robotic hands and a full humanoid robot. Fully containerized with Docker for platform-independent execution.

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Meet our team (Team 2)

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Problem Statement

Platform Compatibility

ROS 2 typically limited by platform issues, especially on macOS and windows where native installation is difficult.

Dual-Hand Coordination

Most simulations limited to single-hand movement or lack coordination between multiple components.

Integrated Simulation

Absence of integrated humanoid and hand simulation with coordinated body movements.

Complex Management

Difficulty managing multiple joint movements simultaneously across different components.

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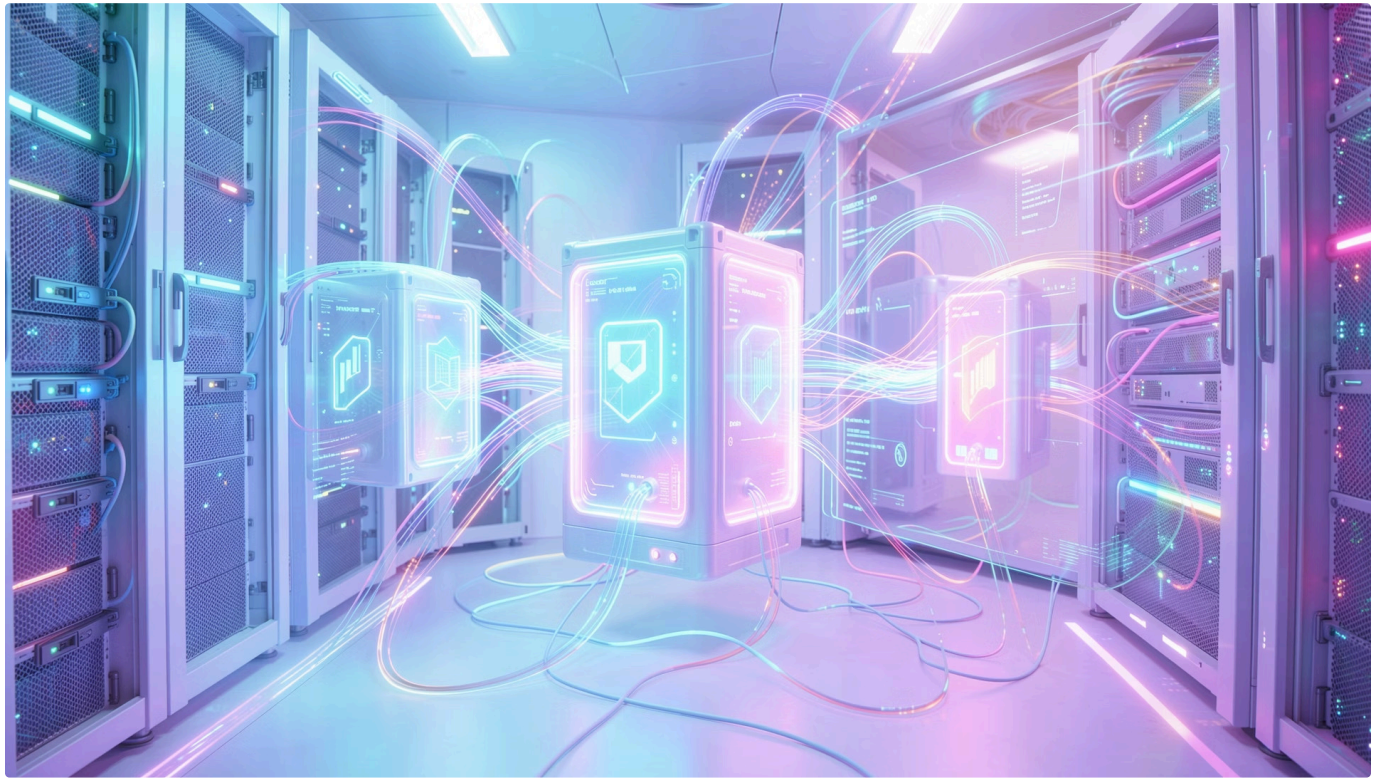
Solution Overview



Docker + VNC Environment

Ensures cross-platform execution on Mac, Windows, and Linux systems.

Dual Robotic Hands

Capable of performing synchronized ASL gestures with coordinated finger movements.

Full Humanoid Model

Integrates body and hand movements for complete gesture simulation.

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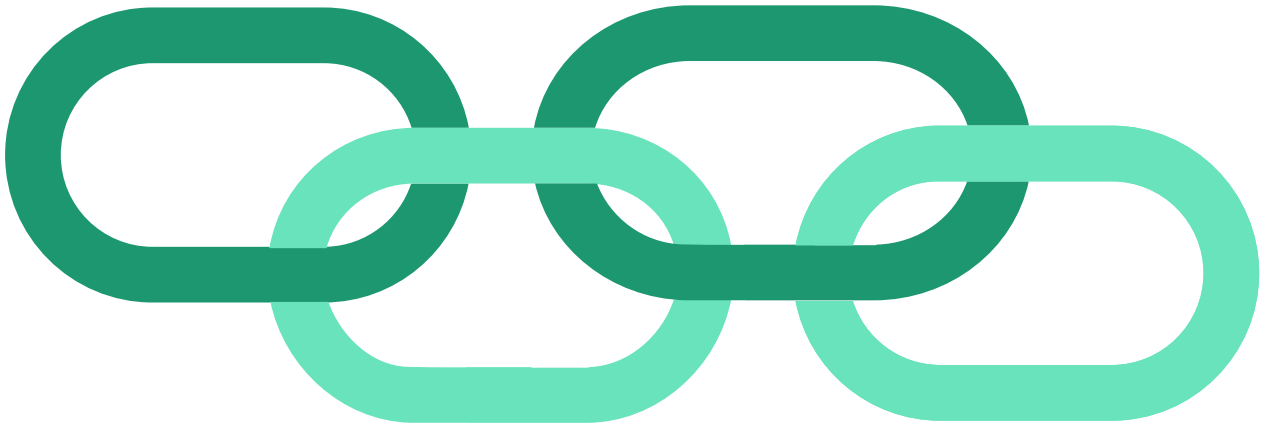
System Architecture



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Docker + VNC Setup

01

Containerized Environment

Docker image includes ROS 2 (Jazzy/Humble), Python dependencies, and VNC server for GUI access.

02

Build & Run

Build Docker image, run container with port mapping, and connect using VNC viewer.

03

Launch RViz2

Launch RViz2 visualization tool inside the container for real-time rendering.

04

Platform Independence

Users run simulation without installing ROS locally, solving compatibility issues.

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Dual Robotic Hands

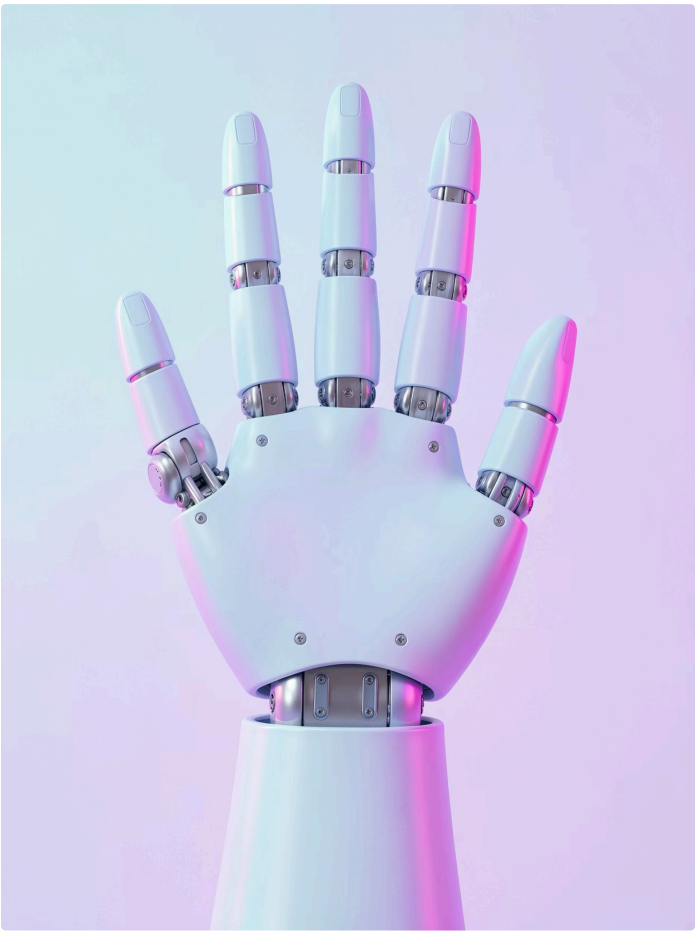
URDF Xacro Design

Two hands placed symmetrically in world frame, each with 14 joints:

- Thumb: 2 joints
- Each finger: 3 joints
- Modeled as chain of links

Joint Movement

Base joints rotate along Z-axis for side movement. Finger joints rotate along Y-axis for bending motion.



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Dual Hands Control Mechanism

1

/finger_count Topic

Accepts 0-10 values. System automatically splits count between left and right hands.

/letter_command Topic

String input for ASL letters. Single letter: both hands same gesture. Two letters: different gestures.

Coordinator Node

Processes input and sends commands to individual hand controllers for joint angle calculation.

Example: Input /finger_count = 8 produces Left hand = 5, Right hand = 3

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Humanoid Robot Overview

1.5m

Height

Full-body simulation model

38

Total Joints

10 body + 28 finger joints

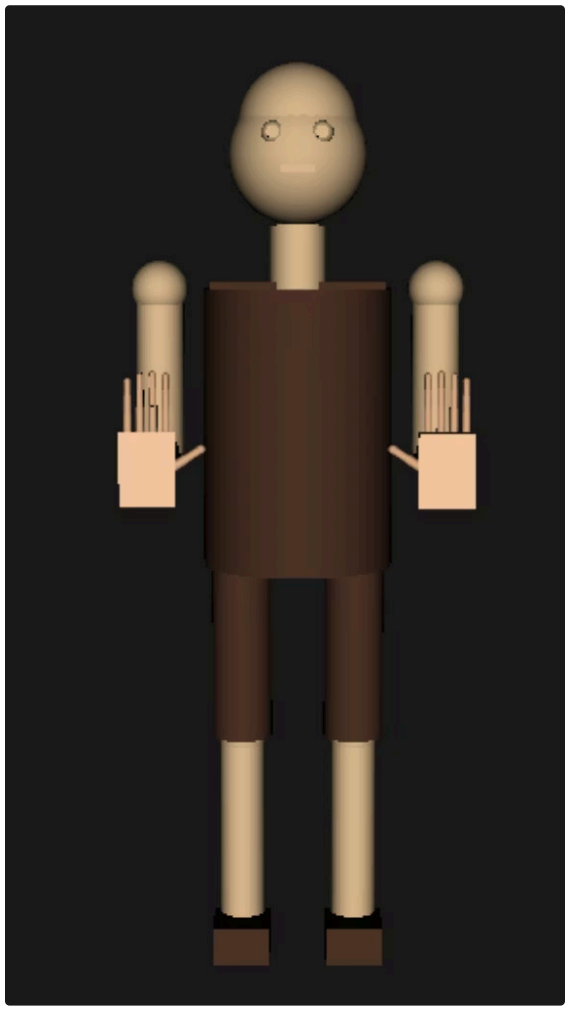
10Hz

Update Rate

Smooth motion control

Body Joints Include:

- Neck pitch and yaw for looking toward a person or following the conversation
- Shoulder pitch and roll to shape arm position and keep signs clear
- Elbow pitch to extend, bend, and refine gesture reach
- Wrist pitch to fine-tune hand orientation for precise ASL forms



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Humanoid Control System

/sign_command Topic

Accepts letters (A–Z), numbers (1–10), words (hello, bye, please, yes), and sequences (alphabet, dance routines).

1

Sign Coordinator

Processes input and manages command execution

2

Body Controller

Controls upper-body movements

3

Hand Controllers

Left and right hand coordination

4

Joint State Publisher

Merges and publishes all states

Uses threading and locking mechanisms for sequential command execution.

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Results & Future Scope

Key Achievements

- 38 active joints with real-time ROS 2 communication
- 62+ ASL gestures supported
- Smooth motion control at 10 Hz
- Cross-platform compatibility using Docker
- Stable, scalable, and efficient system

Future Improvements

- Additional sign languages
- Speech-to-sign conversion
- Mobile app interface
- Real robotic hardware integration
- AI-based gesture recognition

This project demonstrates a complete robotic simulation system for ASL using dual hands and humanoid robot, with applications in assistive technology, robotics research, and human-robot interaction.

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Live Interactive Demo

Explore the robotics simulation in action and watch the demo yourself.

Open the demo: https://ravali0423.github.io/robotics/

See the ASL robotics experience, interact with the system, and get a hands-on look at the project.

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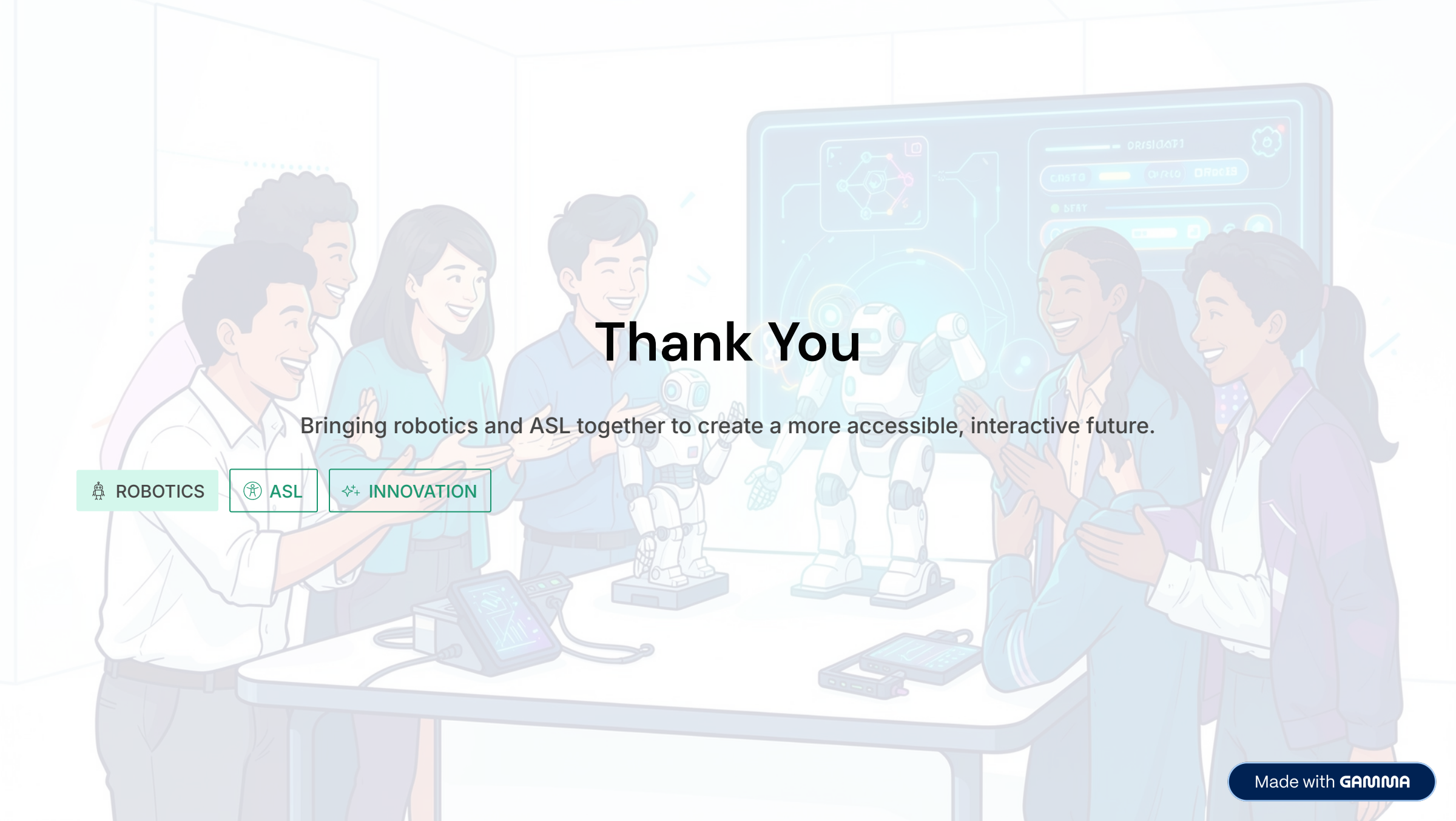

Thank You

Bringing robotics and ASL together to create a more accessible, interactive future.

ROBOTICS | ASL | INNOVATION

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